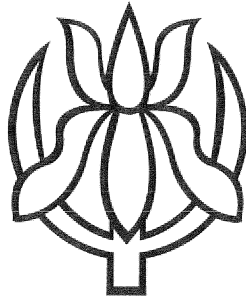


FRENSHAM



YEAR 12 TRIAL HSC EXAMINATION
2011
MATHEMATICS
EXTENSION 2

Time Allowed 3 hours +5 minutes reading time

INSTRUCTIONS:

- All questions may be attempted
- All questions are of equal value
- Show all necessary working. Marks may be deducted for careless or badly arranged work
- Start each question on a new page
- Board of Studies approved calculators may be used

Student name / number

Question 1**Begin a new booklet****Marks**

(a) Find $\int \frac{x^2+1}{\sqrt{x}} dx$. 2

(b) Find $\int \frac{\cos^3 x}{\sin^2 x} dx$ using the substitution $u = \sin x$. 3

(c) Evaluate $\int_0^{\frac{1}{2}\log_e 3} \frac{1}{e^x + e^{-x}} dx$ using the substitution $u = e^x$. 3

(d) Evaluate in simplest exact form $\int_1^e x^3 \log_e x dx$. 3

(e)(i) Using the substitution $t = \tan \frac{x}{2}$, show that 2

$$\int_0^{\frac{\pi}{2}} \frac{1}{5+5\sin x-3\cos x} dx = \int_0^1 \frac{1}{4t^2+5t+1} dt.$$

(ii) Hence evaluate in simplest exact form $\int_0^{\frac{\pi}{2}} \frac{1}{5+5\sin x-3\cos x} dx$. 2

Student name / number

Question 2**Begin a new booklet****Marks**(a) If $z_1 = 2i$ and $z_2 = 1 + 3i$, express in the form $a + ib$ (where a and b are real)

(i) $z_1 + \bar{z}_2$.

1

(ii) $z_1 z_2$.

1

(iii) $\frac{1}{z_2}$.

1

(b)(i) Express $z = 1 + i\sqrt{3}$ in modulus/argument form.

2

(ii) Hence show that $z^{10} + 512z = 0$.

2

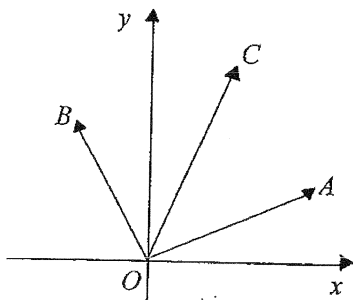
(c)(i) On an Argand diagram sketch the locus of the point P representing z such that $|z - (\sqrt{3} + i)| = 1$.

2

(ii) Find the set of possible values of $|z|$ and the set of possible principal values of $\arg z$.

2

(d)



In the Argand diagram above, vectors $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$ represent the complex numbers z_1 , z_2 and $z_1 + z_2$ respectively, where $z_1 = \cos\theta + i\sin\theta$ and $z_1 + z_2 = (1 + i)z_1$.

(i) Express z_2 in terms of z_1 and show that $OACB$ is a square.

2

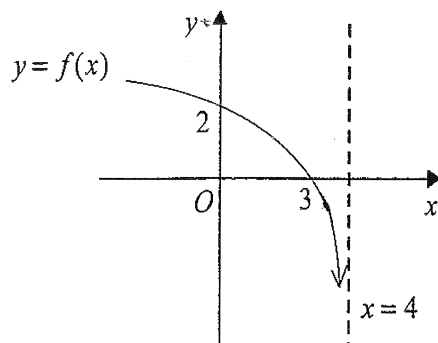
(ii) Show that $(z_1 + z_2)\overline{(z_1 - z_2)} = 2i$.

2

Student name / number

Marks**Question 3****Begin a new booklet**

- (a) The diagram shows the graph of the curve $y = f(x)$. On separate diagrams, sketch the graphs of the curves listed below, showing clearly intercepts on the coordinate axes and the equations of any asymptotes:



(i) $y = |f(x)|$ 1

(ii) $y = f(|x|)$ 1

(iii) $y = f(x^2)$ 2

(iv) $y = \frac{1}{f(x)}$ 2

- (b) $P(x)$ is an even polynomial. Show that when $P(x)$ is divided by $(x^2 - a^2)$, where $a \neq 0$, the remainder is independent of x . 3

- (c) The zeroes of $x^3 + px^2 + qx + r$ are α, β and γ (where p, q and r are real numbers).

(i) Find $\alpha\beta + \alpha\gamma + \beta\gamma$. 1

(ii) Find $\alpha^2 + \beta^2 + \gamma^2$. 1

(iii) Find a cubic polynomial with integer coefficients whose zeroes are $2\alpha, 2\beta$ and 2γ . 2

- (d) If $p > 0$, and $q > 0$, and $p + q = 1$, show that $\frac{1}{p} + \frac{1}{q} \geq 4$. 2

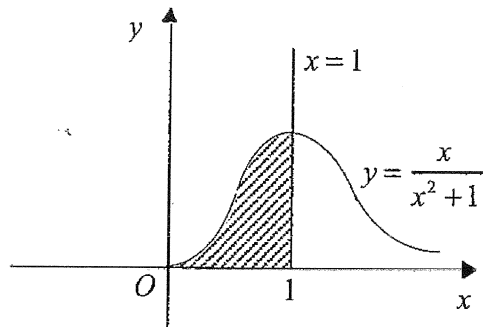
Student name / number

Marks

Question 4

Begin a new booklet

(a)



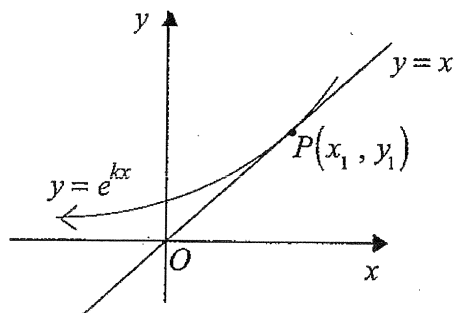
The region bounded by the curve $y = \frac{x}{x^2 + 1}$ and the x -axis between $x = 0$ and $x = 1$ is rotated through one complete revolution about the y -axis.

- (i) Use the method of cylindrical shells to show that the volume V of the solid 1

formed is given by $V = 2\pi \int_0^1 \frac{x^2}{x^2 + 1} dx$

- (ii) Hence find the value of V in simplest exact form. 3

(b)



The line $y = x$ is tangent to the curve $y = e^{kx}$ (where $k > 0$) at the point $P(x_1, y_1)$ on the curve. By considering the gradient of OP show that $k = \frac{1}{e}$. 3

Question 4 continued

(c) The Hyperbola H has the equation $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

- | | | |
|-------|---|----------|
| (i) | Find the eccentricity of H. | 1 |
| (ii) | Find the co-ordinates of the foci of H. | 1 |
| (iii) | Draw a neat one third of a page sketch of H. | 2 |
| (iv) | The line $x = 6$ cuts H at A and B. Find the coordinates of A and B
if A is in the first quadrant. | 2 |
| (v) | Derive the equation of the tangent to H at A. | 2 |

Student name / number

Marks**Question 5****Begin a new booklet**

- (a) A lifebelt mould is made by rotating the circle $x^2 + y^2 = 64$ through one complete revolution about the line $x = 28$, where All the measurements are in centimetres.
- (i) Use the method of slicing to show that the volume, $V \text{ cm}^3$ of the lifebelt is given by
- $$V = 112\pi \int_{-8}^8 \sqrt{64 - y^2} dy.$$
- (ii) Find the exact volume of the lifebelt.
- (b) (i) Show that the tangent to the ellipse $\frac{x^2}{12} + \frac{y^2}{4} = 1$ at the point $P(3, 1)$ has equation $x + y = 4$.
- ii) If this tangent cuts the directrix in the fourth quadrant at the point T , and S is the corresponding focus, show that SP and ST are perpendicular.

Student name / number

Question 6	Begin a new booklet	Marks
(a) i) Show that $\tan(A + \frac{\pi}{2}) = -\cot A$.		2
ii) Use the method of Mathematical Induction, and the result in (i), to show that		4
$\tan\left\{(2n+1)\frac{\pi}{4}\right\} = (-1)^n \quad \text{for all integers } n \geq 1.$		
(b) Given the equation $y^2 + xy + x^2 = 1$		
i) Make y the subject.		2
ii) Hence, <u>or otherwise</u> , find $\frac{dy}{dx}$		2
(c) Given that $z = \cos \theta + i \sin \theta$ and $z^n + z^{-n} = 2 \cos n\theta$, show that		4
$\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$		
(d) Show that $\cos\left(x + \frac{\pi}{2}\right) = -\sin x$.		1

Student name / number

Marks

Question 7

Begin a new booklet

(a)(i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. 2

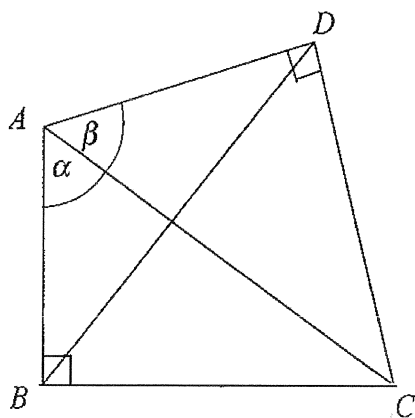
(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sqrt{1 + (x - \frac{\pi}{4})^2}} dx$. 3

(b) Let $I_n = \int_0^1 (1-x^r)^n dx$, where $r > 0$, for $n = 0, 1, 2, \dots$.

(i) Show that $I_n = \frac{nr}{nr+1} I_{n-1}$ for $n = 1, 2, 3, \dots$. 3

(ii) Hence evaluate $\int_0^1 (1-x^{\frac{1}{2}})^3 dx$. 2

(c)



$ABCD$ is a quadrilateral in which $\angle ABC = \angle ADC = \frac{\pi}{2}$, $\angle CAB = \alpha$, $\angle CAD = \beta$ and $AC = 1$.

(i) Show that $\angle BDC = \alpha$. 2

(ii) Hence show that $BD = \sin(\alpha + \beta)$. 3

Student name / number

Marks

Question 8

Begin a new booklet

- a) i) Write the general solution to $\tan 4\theta = 1$ 1

- ii) Use De Moivre's Theorem and the binomial theorem to 3
show that $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$

- iii) Hence find the roots of $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ in 3
the form $x = \tan \theta$.

- iv) Hence prove that : 2
 $\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16} = 28$
(Hint: Let the roots be α, β, γ and δ).

- (b) (i) Use a diagram to explain why 1

$$\int_0^b \sin x \, dx = \lim_{n \rightarrow \infty} \left(\sin \frac{b}{n} + \sin \frac{2b}{n} + \dots + \sin \frac{nb}{n} \right) \cdot \frac{b}{n}$$

$$\text{for } b = \frac{\pi}{2}.$$

- (ii) Given that $2 \sin \theta \sin \alpha = \cos(\theta - \alpha) - \cos(\theta + \alpha)$, show that 2

$$\sum_{k=1}^n \sin\left(\frac{kb}{n}\right) = \frac{\cos\left(\frac{b}{2n}\right) - \cos\left(b + \frac{b}{2n}\right)}{2 \sin\left(\frac{b}{2n}\right)}$$

- (iii) Hence show that $\int_0^b \sin x \, dx = 1 - \cos b$. 3

END OF EXAM

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

①

Frensham 2011 Extension 2 Trial HSC

QUESTION 1

$$\begin{aligned}
 (a) \quad & \int \frac{x^2+1}{x^{\frac{3}{2}}} dx \\
 &= \int x^{\frac{3}{2}} + x^{-\frac{1}{2}} dx \\
 &= \frac{2x^{\frac{5}{2}}}{5} + 2x^{\frac{1}{2}} + C \\
 &= \frac{2\sqrt{x^5}}{5} + 2\sqrt{x} + C
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \int \frac{\cos^2 x \cdot \cos x}{\sin^2 x} dx \\
 &= \int \frac{(1 - \sin^2 x) \cos x}{\sin^2 x} dx \\
 &= \int \frac{1 - u^2}{u^2} du \\
 &= \int u^{-2} - 1 du
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } u &= \sin x \\
 du &= \cos x dx
 \end{aligned}$$

$$\begin{aligned}
 &= -u^{-1} - u + C \\
 &= \frac{-1}{\sin x} - \sin x + C
 \end{aligned}$$

$$\text{OR} \quad -\operatorname{cosec} x - \sin x + C$$

$$(c) \quad \int_0^{\frac{1}{2} \ln 3} \frac{1}{e^x + e^{-x}} dx \times \frac{e^x}{e^x}$$

$$\begin{aligned}
 \text{Let } u &= e^x \\
 du &= e^x dx
 \end{aligned}$$

$$= \int_0^{\ln \sqrt{3}} \frac{e^x dx}{e^{2x} + 1}$$

$$\begin{aligned}
 \text{When } x=0 \quad u &= 1 \\
 x=\ln \sqrt{3} \quad u &= \sqrt{3}
 \end{aligned}$$

$$\begin{aligned}
 &= \int_1^{\sqrt{3}} \frac{du}{u^2 + 1} = \left[\tan^{-1} u \right]_1^{\sqrt{3}} = \tan^{-1} \sqrt{3} - \tan^{-1} 1 \\
 &= \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}
 \end{aligned}$$

(2)

$$(d) \int_1^e x^3 \ln x \, dx$$

$$\text{Let } u = \ln x \quad v = \frac{x^4}{4}$$

$$u' = \frac{1}{x} \quad v' = x^3$$

$$= \left[\frac{x^4}{4} \ln x \right]_1^e - \int_1^e \frac{x^4}{4} \cdot \frac{1}{x} \, dx$$

$$= \frac{e^4}{4} - \int_1^e \frac{x^3}{4} \, dx$$

$$= \frac{e^4}{4} - \left[\frac{x^4}{16} \right]_1^e$$

$$= \frac{e^4}{4} - \left(\frac{e^4}{16} - \frac{1}{16} \right)$$

$$= \frac{3e^4 + 1}{16}$$

$$(e) i) \text{ let } \tan \frac{x}{2} = t$$

$$\sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\int_0^{\pi/2} \frac{dx}{5 + 5\sin x - 3\cos x}$$

$$x = 0, t = 0$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$x = \pi/2, t = 1$$

$$dx = \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{\frac{2}{1+t^2} dt}{5 + \frac{10t}{1+t^2} - \frac{3-3t^2}{1+t^2}}$$

$$= \int_0^1 \frac{2 dt}{5(1+t^2) + 10t - 3 + 3t^2}$$

$$= \int_0^1 \frac{2 dt}{2 + 8t^2 + 10t}$$

$$= \int_0^1 \frac{dt}{1 + 5t + 4t^2}$$

$$= \int_0^1 \frac{dt}{(4t+1)(t+1)}$$

(3)

$$= \int_0^1 \frac{dt}{(4t+1)(t+1)}$$

using partial fractions let $\frac{1}{(4t+1)(t+1)} = \frac{A}{(4t+1)} + \frac{B}{t+1}$

$$A(t+1) + B(4t+1) = 1$$

$$At + A + 4Bt + B = 1$$

$$\therefore At + 4Bt = 0$$

$$A + 4B = 0 \quad \dots \textcircled{1}$$

$$A + B = 1 \quad \dots \textcircled{2}$$

$$\underline{3B = -1} \quad \textcircled{1} - \textcircled{2}$$

$$B = -\frac{1}{3}, \quad A = \frac{4}{3}$$

$$= \int_0^1 \frac{\frac{4}{3}}{4t+1} - \frac{\frac{1}{3}}{t+1} dt$$

$$= \frac{1}{3} \int_0^1 \frac{4}{4t+1} - \frac{1}{t+1} dt$$

$$= \frac{1}{3} \left[\ln(4t+1) - \ln(t+1) \right]_0^1$$

$$= \frac{1}{3} \ln \left(\frac{4t+1}{t+1} \right) \Big|_0^1$$

$$= \frac{1}{3} \left[\ln \left(\frac{5}{2} \right) - \ln 1 \right]$$

$$= \frac{1}{3} \ln \frac{5}{2}$$

4

QUESTION 2

$$a) z_1 = 2i, z_2 = 1+3i, \bar{z}_2 = 1-3i$$

$$i) z_1 + \bar{z}_2 = 2i + 1 - 3i \\ = 1 - i$$

✓

$$ii) z_1 z_2 = 2i(1+3i) \\ = 2i - 6 \\ = -6 + 2i$$

✓

$$iii) \frac{1}{z_2} = \frac{1}{1+3i} \times \frac{1-3i}{1-3i}$$

$$= \frac{1-3i}{1+9}$$

$$= \frac{1-3i}{10} = \frac{1}{10} - \frac{3}{10}i$$

✓

$$b) z = 1 + i\sqrt{3} \quad |z| = \sqrt{1^2 + \sqrt{3}^2} = 2 \\ \theta = \tan^{-1} \sqrt{3} = \pi/3$$

$$z = 2(\cos \pi/3 + i \sin \pi/3)$$

✓ Modulus
✓ argument

$$ii) z^{10} + 512z = 2^{10}(\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3}) + 2^{10}(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$$

$$= 2^{10}(\cos(2\pi + \frac{4\pi}{3}) + i \sin(2\pi + \frac{4\pi}{3}) + \cos(\frac{\pi}{3}) + i \sin(\frac{\pi}{3}))$$

$$= 2^{10}(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} + \cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$$

$$= 2^{10}(-\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} + \cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$$

$$= 0$$

①

①

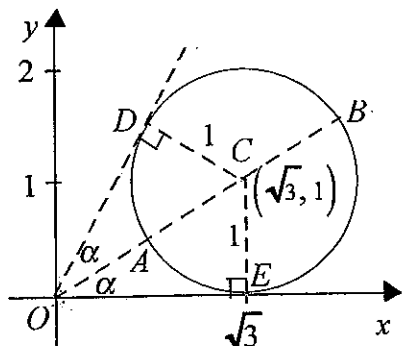
c. Outcomes assessed : E3

Marking Guidelines

Criteria	Marks
i • sketches a circle with correct centre	1
• sketches a circle with correct radius	1
ii • states set of values for $ z $	1
• states set of values for $\arg z$	1

Answer

i. $|z - (\sqrt{3} + i)| = 1$



$$\alpha = \tan^{-1} \frac{1}{\sqrt{3}}, \quad \arg z = 2\alpha = \frac{2\pi}{6} = \frac{\pi}{3}$$

ii. $OC = 2$ and $\alpha = \frac{\pi}{6}$

$$OA \leq |z| \leq OB \quad \therefore 1 \leq |z| \leq 3$$

$$0 \leq \arg z \leq \angle EOD \quad \therefore 0 \leq \arg z \leq \frac{\pi}{3}$$

d. Outcomes assessed : E3

Marking Guidelines

Criteria	Marks
i • expresses z_2 in terms of z_1	1
• explains why $OACB$ is a square	1
ii • uses properties of a square to deduce $z_1 + z_2 = i(z_1 - z_2)$	1
• uses the side and diagonal lengths of the square to complete the proof	1

Answer

$$z_2 = z_1 + iz_1 - z_1$$

i. $z_1 + z_2 = (1+i)z_1 \quad \therefore z_2 = iz_1$ Hence OB is the rotation of OA anticlockwise by 90° .
Hence $OACB$ is a parallelogram in which $OA = OB$ and $\angle AOB = 90^\circ$. $\therefore OACB$ is a square.

ii. The diagonals of a square are equal and meet at right angles.

$\therefore OC$ is the anticlockwise rotation of BA by 90° . Hence

$$z_1 + z_2 = i(z_1 - z_2)$$

$$(z_1 + z_2)\overline{(z_1 - z_2)} = i|z_1 - z_2|^2$$

But $BA^2 = OA^2 + OB^2 = 1 + 1 \Rightarrow |z_1 - z_2|^2 = 2$.

$$\therefore (z_1 + z_2)\overline{(z_1 - z_2)} = 2i$$

Question 3

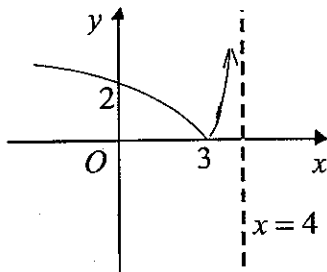
a. Outcomes assessed : E6

Marking Guidelines

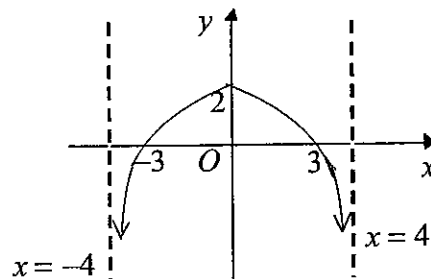
Criteria	Marks
i • copies curve for $x \leq 3$ and reflects section of curve for $x > 3$ in x -axis	1
ii • copies curve for $x \geq 0$ and includes reflection of this section of curve in the y -axis	1
iii • sketches curve that is concave down, symmetric in the y -axis, with turning point $(0,2)$	1
• shows asymptotes and x -intercepts	1
iv • shows vertical asymptote $x = 3$ and sketches left hand branch correctly	1
• sketches right hand branch correctly showing nature at $x = 4$	1

Answer

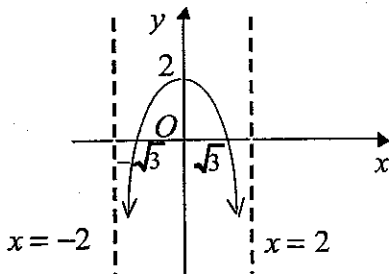
i. $y = |f(x)|$



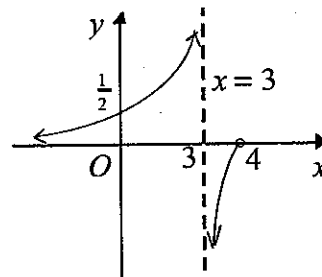
ii. $y = f(|x|)$



iii. $y = f(x^2)$



iv. $y = \frac{1}{f(x)}$



b. Outcomes assessed : E4

Marking Guidelines

Criteria	Marks
• states remainder on division by $(x^2 - a^2)$ is $(cx + d)$ for some constants c and d	1
• uses definition of an even function to deduce $ca + d = -ca + d$	1
• completes proof by showing $c = 0$	1

Answer

$P(x) \equiv (x^2 - a^2)Q(x) + cx + d$ for constants c, d where $cx + d$ is the remainder on division by $x^2 - a^2$.

$P(x)$ even $\Rightarrow P(-a) = P(a) \quad \therefore ca + d = -ca + d$

$$2ca = 0$$

But $a \neq 0 \quad \therefore c = 0$.

Hence remainder is some constant d , which is independent of x .

(7)

(Q3)

c) $x^3 + px^2 + qx + r$ has zeroes α, β, γ

$$i) \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = q \quad (1)$$

$$ii) \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ = (-p)^2 - 2(q) \\ = p^2 - 2q \quad (1)$$

(iii) Let $x = 2y$ $y = \frac{x}{2}$ sub into poly

$$\left(\frac{x}{2}\right)^3 + p\left(\frac{x}{2}\right)^2 + q\left(\frac{x}{2}\right) + r = 0 \quad (1)$$

$$\frac{x^3}{8} + \frac{px^2}{4} + \frac{qx}{2} + r = 0$$

$$x^3 + 2px^2 + 4qx + 8r = 0 \quad (1)$$

(d) $p > 0, q > 0, p+q=1$

$$\frac{p+q}{2} \geq \sqrt{pq} \quad (\text{arithmetic mean} \geq \text{geometric mean})$$

$$p+q \geq 2\sqrt{pq} \quad \text{Let } p = \frac{1}{p} \text{ \& } q = \frac{1}{q}$$

$$\frac{1}{p} + \frac{1}{q} \geq 2\sqrt{\frac{1}{pq}} \quad \dots \star$$

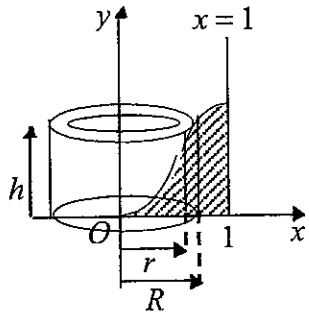
$$\frac{p+q}{pq} \geq \frac{2}{\sqrt{pq}}$$

$$\frac{pq}{p+q} \leq \frac{\sqrt{pq}}{2} \quad \text{divide both sides by } \sqrt{pq} \text{ \& multiply by } (p+q)$$

$$\frac{\sqrt{pq}}{\sqrt{pq}} \times \frac{pq}{\sqrt{pq}} \leq \frac{p+q}{2} \quad \text{But } p+q=1 \\ \sqrt{pq} \leq \frac{p+q}{2} \quad \text{so } \frac{1}{\sqrt{pq}} \geq 2 \\ \text{so } \frac{1}{p} + \frac{1}{q} \geq 4 \text{ from } \star$$

Answer

i.



$$h = \frac{x}{x^2 + 1}$$

$$r = x$$

$$R = x + \delta x$$

$$\begin{aligned}\delta V &= \pi(R^2 - r^2)h \\ &= \pi(R + r)(R - r)h \\ &= \pi(2x + \delta x)(\delta x) \frac{x}{x^2 + 1}\end{aligned}$$

Ignoring terms in $(\delta x)^2$,

$$\begin{aligned}V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{x=1} \pi \frac{2x^2}{x^2 + 1} \delta x \\ &= 2\pi \int_0^1 \frac{x^2}{x^2 + 1} dx\end{aligned}$$

$$\text{ii. } V = 2\pi \int_0^1 \left(1 - \frac{1}{x^2 + 1}\right) dx = 2\pi \left[x - \tan^{-1} x\right]_0^1 \quad \therefore V = \frac{\pi}{2}(4 - \pi)$$

b. Outcomes assessed : E6

Marking Guidelines

Criteria	Marks
• differentiates to obtain gradient of tangent at P	1
• uses gradient of OP is 1 to deduce $x_1 = y_1 = \frac{1}{k}$	1
• substitutes in equation of curve to find k .	1

Answer

$$y = e^{kx} \quad \therefore \frac{dy}{dx} = ke^{kx} \quad \text{Hence tangent at } P \text{ has gradient } ke^{kx_1} = ky_1, \text{ since } y_1 = e^{kx_1}.$$

$$\text{But gradient of } OP \text{ is } 1 \text{ (since } P \text{ lies on line } y = x) \quad \therefore ky_1 = 1 \quad \text{and hence } x_1 = y_1 = \frac{1}{k}.$$

$$\text{Then since } P \text{ lies on } y = e^{kx}, \quad \frac{1}{k} = e^{k \cdot \frac{1}{k}} \quad \therefore k = \frac{1}{e}.$$

c. Outcomes assessed : E3, E4

Marking Guidelines

Criteria	Marks
i* • uses O, P, Q collinear to deduce result	1
ii • writes the coordinates of two of the points	1
• writes the coordinates of the remaining two points	1
iii • deduces that $XYUV$ is a rhombus	1
• expresses the area of the rhombus in terms of its diagonal lengths to obtain the result	1
iv • compares the areas of the quadrilateral and ellipse to deduce that $ \sin 2\theta = 1$	1
• states the four values of θ	1
• sketches the ellipse inscribed in the quadrilateral giving the required detail.	1

9

4(c) $\frac{x^2}{25} - \frac{y^2}{9} = 1$

i)

$a = 5, b = 3$

$b^2 = a^2(e^2 - 1)$

$9 = 25(e^2 - 1)$

$e^2 - 1 = \frac{9}{25}$

$e^2 = \frac{34}{25}$

$e = \frac{\sqrt{34}}{5}$

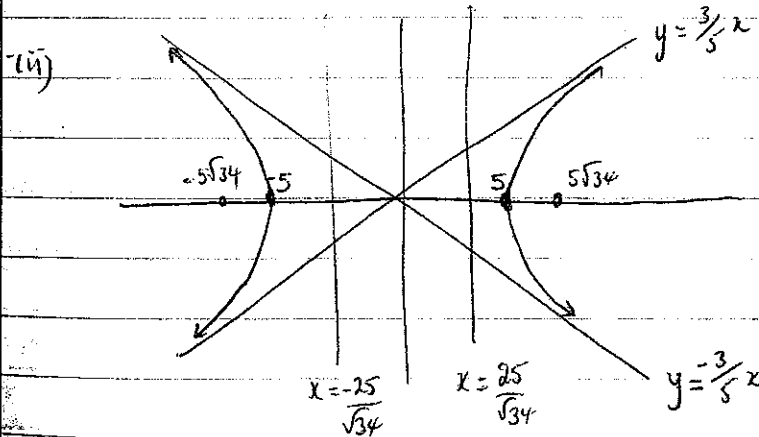
①

ii) foci $(\pm ae, 0)$

$(\pm 5 \times \frac{\sqrt{34}}{5}, 0)$

$(\pm \sqrt{34}, 0)$

①



iv) $\frac{6^2}{25} - \frac{y^2}{9} = 1$

$324 - 25y^2 = 225$

$25y^2 = 99$

$y^2 = \frac{99}{25} \therefore y = \pm \frac{\sqrt{99}}{5}$

$\therefore A \text{ is } (6, \frac{\sqrt{99}}{5}), B \text{ is } (6, -\frac{\sqrt{99}}{5})$

v) $\frac{x^2}{25} - \frac{y^2}{9} = 1$

$\frac{2x}{25} - \frac{2y \cdot \frac{dy}{dx}}{9} = 0$

(10)

$$\therefore \frac{dy}{dx} = \frac{-2x}{25} \cdot \frac{-9}{2y}$$
$$= \frac{9x}{25y}$$

$$\text{at } \left(6, \frac{\sqrt{99}}{5}\right) \quad \frac{dy}{dx} = \frac{9}{25} \cdot 6 \cdot \frac{5}{\sqrt{99}}$$

$$= \frac{54}{5\sqrt{99}}$$

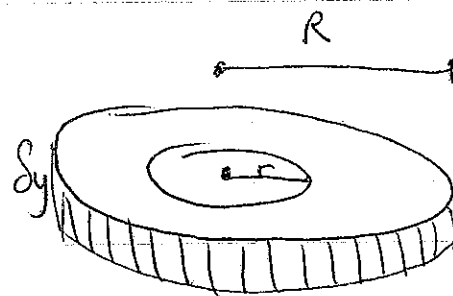
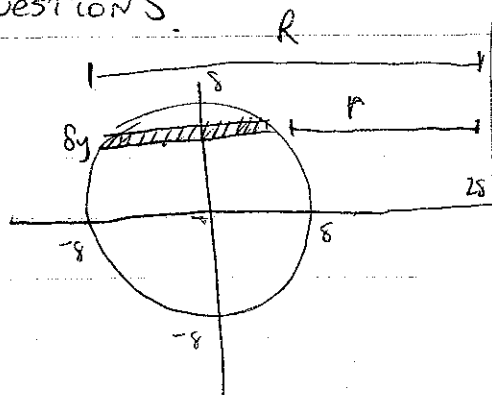
$$\therefore y - \frac{\sqrt{99}}{5} = \frac{54}{5\sqrt{99}} (x - 6)$$

$$5\sqrt{99} y - 99 = 54x - 324$$

$$54x - 5\sqrt{99} y - 225 = 0$$

QUESTIONS

(a) 1)



$$R = 28 + \sqrt{64 - y^2}$$

$$r = 28 - \sqrt{64 - y^2}$$

The volume of the Slice is $\delta V = \pi (R^2 - r^2) \delta y$

$$= \pi (R + r)(R - r) \delta y$$

$$= \pi (56) 2\sqrt{64 - y^2} \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=-8}^8 112\pi \sqrt{64 - y^2} \delta y$$

$$= 112\pi \int_{-8}^8 \sqrt{64 - y^2} dy$$

11) $\int_{-8}^8 \sqrt{64 - y^2} dy$ is a Semi Circle Radius 8

Area = $\frac{1}{2} \times \pi \times r^2$

$$= \frac{1}{2} \times \pi \times 8^2 = 32\pi$$

$$\therefore V = 112\pi \times 32\pi$$

$$= 3584\pi^2 \text{ cm}^3$$

11

5

$$(b) \quad \frac{x^2}{12} + \frac{y^2}{4} = 1$$

(12)

$$\frac{2x}{12} + \frac{2y}{4} \frac{dy}{dx} = 0 \quad \text{using implicit diff.}$$

$$\frac{x}{6} + y \cdot \frac{dy}{dx} = 0$$

$$\frac{x}{6} = -y \cdot \frac{dy}{dx}$$

$$2x = -6y \frac{dy}{dx}$$

$$\frac{dy}{dx} = -\frac{x}{3y} \quad \text{at } P(3, 1)$$

$$m = \frac{dy}{dx} = -\frac{3}{3} = -1$$

$$\text{Tangent is } y - y_1 = m(x - x_1)$$

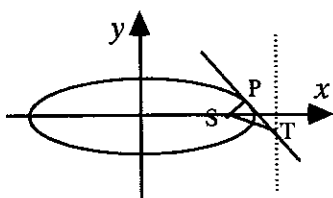
$$y - 1 = -1(x - 3)$$

$$y - 1 = -x + 3$$

$$x + y - 4 = 0$$

$$x + y = 4$$

(ii) If this tangent cuts the directrix in the fourth quadrant at the point T, and S is the corresponding focus, show that SP and ST are perpendicular.



$$\frac{b^2}{a^2} = 1 - e^2 \quad \therefore e = \sqrt{\frac{2}{3}}$$

$$\therefore \text{directrix is } x = \frac{a}{e} \rightarrow x = 3\sqrt{2}$$

$$\text{and focus at } x = ae \rightarrow S(2\sqrt{2}, 0)$$

Putting $x = 3\sqrt{2}$ in $x + y = 4$ gives...

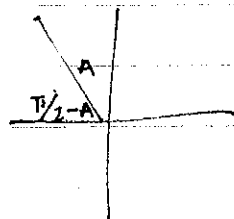
$$T(3\sqrt{2}, 4 - 3\sqrt{2})$$

$$\text{Now, } m_{SP} = \frac{1}{3 - 2\sqrt{2}} \quad \text{and} \quad m_{ST} = \frac{4 - 3\sqrt{2}}{\sqrt{2}}$$

$$m_{SP} \times m_{ST} = \frac{1}{3 - 2\sqrt{2}} \times \frac{4 - 3\sqrt{2}}{\sqrt{2}} = \frac{4 - 3\sqrt{2}}{3\sqrt{2} - 4} = -1 \quad \therefore PS \perp ST.$$

Question 6

$$\begin{aligned}
 \text{a) i) } \tan(A + \pi/2) &= \tan(\pi - (\pi/2 - A)) \\
 &= -\tan(\pi/2 - A) \\
 &= -\cot A
 \end{aligned}$$



$$\text{ii) Aim to prove } \tan\left\{(2n+1)\frac{\pi}{4}\right\} = (-1)^n \quad \forall \mathbb{Z}^+$$

Step 1 Prove true for $n=1$

$$\text{LHS} = \tan\left\{(2(1)+1)\frac{\pi}{4}\right\}$$

$$\text{RHS} = (-1)^1 = -1$$

$$= \tan \frac{3\pi}{4}$$

$$= -\tan \pi/4$$

$$= -1$$

$\therefore \text{LHS} = \text{RHS}$ True for $n=1$

Step 2 Assume true for $n=k$

$$\text{Assume } \tan\left\{(2k+1)\frac{\pi}{4}\right\} = (-1)^k \quad \dots (*)$$

Step 3 Prove true for $n=k+1$

$$\text{Prove } \tan\left\{(2(k+1)+1)\frac{\pi}{4}\right\} = (-1)^{k+1}$$

$$\text{LHS} = \tan\left\{(2(k+1)+1)\frac{\pi}{4}\right\}$$

$$= \tan\left\{(2k+3)\frac{\pi}{4}\right\}$$

$$= \tan\left(2k \cdot \frac{\pi}{4} + \frac{3\pi}{4}\right)$$

$$\frac{3\pi}{4} = \frac{\pi}{2} + \frac{\pi}{4}$$

$$= \frac{\pi}{2} + \frac{\pi}{4}$$

$$= \tan\left(\frac{k\pi}{2} + \frac{\pi}{2} + \frac{\pi}{4}\right)$$

$$= \tan\left(\left(\frac{k\pi}{2} + \frac{\pi}{2}\right) + \frac{\pi}{4}\right)$$

$$= -\cot\left(\frac{k\pi}{2} + \frac{\pi}{4}\right) \text{ using (i)}$$

$$= -\cot\left(\frac{2k\pi + \pi}{4}\right)$$

$$= -\cot\left\{\left(2k+1\right)\frac{\pi}{4}\right\}$$

$$= \frac{-1}{\tan\left\{(2k+1)\left(\frac{\pi}{4}\right)\right\}}$$

$$= \frac{-1}{(-1)^k} \text{ using } \text{---} *$$

$$= (-1)^1 \div (-1)^k$$

$$= (-1)^{1-k}$$

$$= (-1) \times (-1)^{-k}$$

$$= (-1) \times (-1)^k$$

$$= (-1)^{k+1} = \text{RHS}$$

$\therefore$ Statement is true for $n = k+1$.

STEP 4 we have proven Statement is true for $n=1$.

Assuming it is true for $n=k$, we have proven it true for $n=k+1$, $\therefore$ it is true for $n=1+1=2$, $n=2+1=3$, and so on for all integral $n \geq 1$.

Q6

$$b) y^2 + xy + x^2 = 1$$

$$y^2 + xy = 1 - x^2$$

Complete the square

$$y^2 + xy + \left(\frac{x}{2}\right)^2 = 1 - x^2 + \left(\frac{x}{2}\right)^2$$

$$\left(y + \frac{x}{2}\right)^2 = 1 - x^2 + \frac{x^2}{4}$$

$$\left(y + \frac{x}{2}\right)^2 = 1 - \frac{3x^2}{4}$$

$$y = -\frac{x}{2} \pm \sqrt{1 - \frac{3x^2}{4}}$$

$$y = -\frac{x}{2} \pm \frac{\sqrt{4 - 3x^2}}{2}$$

$$y = \frac{-x \pm \sqrt{4 - 3x^2}}{2}$$

$$(ii) y^2 + xy + x^2 = 1$$

$$2y \frac{dy}{dx} + x \frac{dy}{dx} + y + 2x = 0$$

$$\frac{dy}{dx} (2y + x) + y + 2x = 0$$

$$\frac{dy}{dx} (2y + x) = -(y + 2x)$$

$$\frac{dy}{dx} = \frac{-(y + 2x)}{2y + x}$$

$$06c) \quad z = \cos \theta + i \sin \theta \quad z^n + z^{-n} = 2 \cos n\theta$$

$$z^1 + z^{-1} = 2 \cos \theta \quad \therefore (z^1 + z^{-1})^4 = 16 \cos^4 \theta$$

$$\text{Also } (z^1 + z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$$

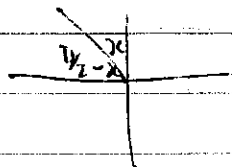
$$= (z^4 + z^{-4}) + 4(z^2 + z^{-2}) + 6$$

$$= 2 \cos 4\theta + 8 \cos 2\theta + 6$$

$$\therefore 16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$$

$$\therefore \cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$$

$$6d) \cos \left(x + \frac{\pi}{2} \right)$$



$$= \cos \left(\pi - \left(\frac{\pi}{2} - x \right) \right)$$

$$= -\cos \left(\frac{\pi}{2} - x \right)$$

$$= -\sin x$$

Question 7

a. Outcomes assessed : E8

Marking Guidelines

Criteria	Marks
i • makes the substitution $u = a - x$	1
• uses property that value of a definite integral does not depend on the variable of integration	1
ii • uses the result from (i) to write the given definite integral with $\cos^2 x$ replacing $\sin^2 x$	1
• uses the table of standard integrals to find the primitive of twice the given integral	1
• evaluates the given integral by substitution of the limits and rearranging	1

Answer

i. Let $u = a - x$
 Then $du = -dx$
 and
 $x = 0 \Rightarrow u = a$
 $x = a \Rightarrow u = 0$

$$\int_0^a f(x) dx = \int_a^0 f(a-u) \cdot -du$$

$$= \int_0^a f(a-u) du$$

$$= \int_0^a f(a-x) dx$$

ii. Let $a = \frac{\pi}{2}$, $f(x) = \frac{\sin^2 x}{\sqrt{1+(x-\frac{\pi}{4})^2}}$. Then $f(\frac{\pi}{2}-x) = \frac{\sin^2(\frac{\pi}{2}-x)}{\sqrt{1+(\frac{\pi}{2}-x-\frac{\pi}{4})^2}} = \frac{\cos^2 x}{\sqrt{1+(x-\frac{\pi}{4})^2}}$.

Using (i), if $I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sqrt{1+(x-\frac{\pi}{4})^2}} dx$, then $I = \int_0^{\frac{\pi}{2}} \frac{\cos^2 x}{\sqrt{1+(x-\frac{\pi}{4})^2}} dx$.

$$2I = \int_0^{\frac{\pi}{2}} \left\{ \frac{\sin^2 x}{\sqrt{1+(x-\frac{\pi}{4})^2}} + \frac{\cos^2 x}{\sqrt{1+(x-\frac{\pi}{4})^2}} \right\} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1+(x-\frac{\pi}{4})^2}} dx$$

$$= \left[\ln \left\{ \left(x - \frac{\pi}{4} \right) + \sqrt{1 + \left(x - \frac{\pi}{4} \right)^2} \right\} \right]_0^{\frac{\pi}{2}}$$

$$= \ln \left\{ \frac{\frac{\pi}{4} + \sqrt{1 + \left(\frac{\pi}{4} \right)^2}}{-\frac{\pi}{4} + \sqrt{1 + \left(\frac{\pi}{4} \right)^2}} \right\}$$

$$\therefore I = \frac{1}{2} \ln \left\{ \frac{\pi + \sqrt{16 + \pi^2}}{-\pi + \sqrt{16 + \pi^2}} \right\}$$

$$= \frac{1}{2} \ln \left\{ \frac{\left(\pi + \sqrt{16 + \pi^2} \right)^2}{(16 + \pi^2) - \pi^2} \right\}$$

$$= \frac{1}{2} \ln \left\{ \frac{1}{4} \left(\pi + \sqrt{16 + \pi^2} \right)^2 \right\}$$

$$= \ln \left\{ \frac{1}{4} \left(\pi + \sqrt{16 + \pi^2} \right) \right\}$$

b. Outcomes assessed : E8

Marking Guidelines

Criteria	Marks
i • applies integration by parts	1
• evaluates the first part and rearranges the second integrand	1
• expresses the second integral in terms of I_n , I_{n-1} then rearranges to obtain result	1
ii • uses the recurrence formula to express I_3 in terms of I_0	1
• evaluates I_0 and hence evaluates I_3	1

Answer

i. $I_n = \int_0^1 (1-x^r)^n dx$, $n=0, 1, 2, \dots$ where $r > 0$

For $n=1, 2, 3, \dots$

$$I_n = \left[x(1-x^r)^n \right]_0^1 - n \int_0^1 x \cdot (1-x^r)^{n-1} \cdot (-rx^{r-1}) dx$$

$$= 0 - nr \int_0^1 \{ (1-x^r) - 1 \} (1-x^r)^{n-1} dx$$

$$= nr \{ -I_n + I_{n-1} \}$$

$$\therefore (nr+1)I_n = nrI_{n-1}$$

$$I_n = \frac{nr}{nr+1} I_{n-1}$$

ii. For $r = \frac{3}{2}$, $I_3 = \frac{(3 \times \frac{3}{2})}{(3 \times \frac{3}{2} + 1)} \cdot \frac{(2 \times \frac{3}{2})}{(2 \times \frac{3}{2} + 1)} \cdot \frac{(1 \times \frac{3}{2})}{(1 \times \frac{3}{2} + 1)} I_0 = \frac{9}{11} \cdot \frac{3}{4} \cdot \frac{3}{5} I_0$

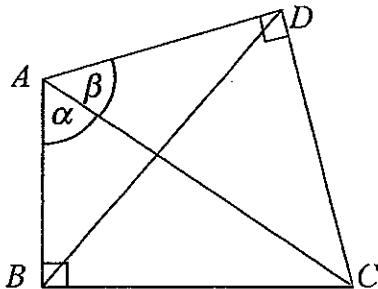
But $I_0 = \int_0^1 1 dx = 1$. Hence $I_3 = \frac{81}{220}$.

c. Outcomes assessed : PE2, PE3

Marking Guidelines

Criteria	Marks
i • explains why ABCD is a cyclic quadrilateral	1
• uses 'angles in the same segment' to deduce result	1
ii • explains why $\sin \angle BCD = \sin(\alpha + \beta)$	1
• explains why $BC = \sin \alpha$	1
• uses the sine rule in $\triangle BCD$ to obtain required result	1

Answer



i. $ABCD$ is a cyclic quadrilateral (opposite angles ABC and ADC are supplementary)
 $\therefore \angle BDC = \angle BAC = \alpha$ (in circle $ABCD$, angles subtended at circumference by same arc BC are equal)

ii. $\angle BCD = \pi - (\alpha + \beta)$ (opposite angles of a cyclic quadrilateral are supplementary)
 $\therefore \sin \angle BCD = \sin \{ \pi - (\alpha + \beta) \} = \sin(\alpha + \beta)$

Also in $\triangle ABC$, $BC = AC \sin \alpha = \sin \alpha$ (given $AC = 1$)

Hence in $\triangle BCD$, $\frac{BD}{\sin \angle BCD} = \frac{BC}{\sin \angle BDC} \Rightarrow \frac{BD}{\sin(\alpha + \beta)} = \frac{\sin \alpha}{\sin \alpha} = 1$. $\therefore BD = \sin(\alpha + \beta)$

Question 8

ai) $\tan 4\theta = 1$

$$\tan 4\theta = \tan \frac{\pi}{4}$$

$$4\theta = \pi n + \frac{\pi}{4}$$

$$\theta = \frac{\pi n + \frac{\pi}{4}}{4}$$

$$= \frac{4\pi n + \pi}{16}$$

✓

ii) $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$ (de Moivre) ✓

By the Binomial Theorem

$$(\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4\cos^3 \theta i \sin \theta + 6\cos^2 \theta i^2 \sin^2 \theta + 4\cos \theta i^3 \sin^3 \theta + i^4 \sin^4 \theta$$

Equating Real + Imaginary Coefficients

$$\sin 4\theta = 4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta$$

$$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$\therefore \tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta} = \frac{4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta}{\cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

dividing by $\frac{\cos^4 \theta}{\cos^4 \theta}$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

(iii) $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$

$$x^4 - 6x^2 + 1 = 4x - 4x^3$$

$$\frac{4x - 4x^3}{x^4 - 6x^2 + 1} = 1$$

Let $x = \tan \theta$ then $\frac{4 \tan \theta - 4 \tan^3 \theta}{\tan^4 \theta - 6 \tan^2 \theta + 1} = 1$

$$\text{i.e. } \tan 4\theta = 1$$

$$\therefore \theta = \frac{\pi n}{4} + \frac{\pi}{16}, n \in \mathbb{Z} \text{ from (1)}$$

Consider $n = 0, \pm 1, \pm 2$

$$\text{i.e. } x = \tan \pi/16, \tan 5\pi/16, -\tan 3\pi/16 \text{ (or } \tan 13\pi/16)$$

$$\text{and } -\tan 7\pi/16 \text{ (or } \tan 9\pi/16)$$

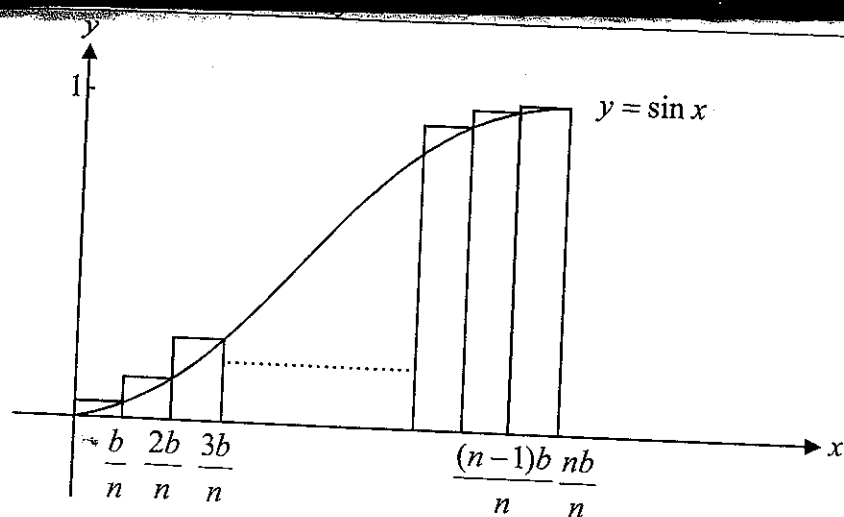
$$(iv) \tan^2 \pi/16 + \tan^2 3\pi/16 + \tan^2 5\pi/16 + \tan^2 7\pi/16$$

$$= (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$$

$$= (-4)^2 - 2(-6)$$

$$= 28 \text{ as required.}$$

(b) (1)



The diagram shows a series of upper rectangles each of width $\frac{b}{n}$ and of height $\sin \frac{b}{n}, \sin \frac{2b}{n}, \sin \frac{3b}{n}, \dots, \sin \frac{nb}{n}$ respectively as you move from left to right.

The sum of the area of the rectangles is $\left(\sin \frac{b}{n} + \sin \frac{2b}{n} + \dots + \sin \frac{nb}{n} \right) \cdot \frac{b}{n}$.

The area under the graph of $y = \sin x$ between $x = 0$ and $x = b$ where $b = \frac{\pi}{2}$

is therefore given by $\lim_{n \rightarrow \infty} \left(\sin \frac{b}{n} + \sin \frac{2b}{n} + \dots + \sin \frac{nb}{n} \right) \cdot \frac{b}{n}$.

1 mark	Explanation including diagram
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Question 8 (cont'd)

(ii) Now,

$$\begin{aligned}
 & 2 \sin\left(\frac{b}{2n}\right) \left(\sin\left(\frac{b}{n}\right) + \sin\left(\frac{2b}{n}\right) + \dots + \sin\left(\frac{nb}{n}\right) \right) \\
 &= \cos\left(\frac{b}{2n} - \frac{b}{n}\right) - \cos\left(\frac{b}{2n} + \frac{b}{n}\right) \\
 &+ \cos\left(\frac{b}{2n} - \frac{2b}{n}\right) - \cos\left(\frac{b}{2n} + \frac{2b}{n}\right) \\
 &+ \cos\left(\frac{b}{2n} - \frac{3b}{n}\right) - \cos\left(\frac{b}{2n} + \frac{3b}{n}\right) + \\
 &\quad \cdot \\
 &\quad \cdot \\
 &\quad \cdot \\
 &+ \cos\left(\frac{b}{2n} - \frac{nb}{n}\right) - \cos\left(\frac{b}{2n} + \frac{nb}{n}\right) \\
 &= \cos\left(\frac{b}{2n}\right) - \cos\left(\frac{3b}{2n}\right) \\
 &+ \cos\left(\frac{3b}{2n}\right) - \cos\left(\frac{5b}{2n}\right) \\
 &+ \cos\left(\frac{5b}{2n}\right) - \cos\left(\frac{7b}{2n}\right) + \\
 &\quad \cdot \\
 &\quad \cdot \\
 &\quad \cdot \\
 &+ \cos\left(\frac{b}{2n} - \frac{nb}{n}\right) - \cos\left(\frac{b}{2n} + \frac{nb}{n}\right) \\
 &= \cos\left(\frac{b}{2n}\right) - \cos\left(b + \frac{b}{2n}\right) \\
 \text{So, } & 2 \sin\left(\frac{b}{2n}\right) \sum_{k=1}^n \sin\left(\frac{kb}{n}\right) = \cos\left(\frac{b}{2n}\right) - \cos\left(b + \frac{b}{2n}\right) \\
 \text{So, } & \sum_{k=1}^n \sin\left(\frac{kb}{n}\right) = \frac{\cos\left(\frac{b}{2n}\right) - \cos\left(b + \frac{b}{2n}\right)}{2 \sin\left(\frac{b}{2n}\right)}
 \end{aligned}$$

as required

2 marks	Multiplying $\sum_{k=1}^n \sin\left(\frac{kb}{n}\right)$ by $2 \sin\left(\frac{b}{2n}\right)$ and obtaining correct expression
1 mark	First part only

Question 8 (cont'd)

(iii) We must use what has already been found.

$$\begin{aligned}
 \int_0^b \sin x \, dx &= \lim_{n \rightarrow \infty} \left(\sin\left(\frac{b}{n}\right) + \sin\left(\frac{2b}{n}\right) + \dots + \sin\left(\frac{nb}{n}\right) \right) \cdot \frac{b}{n} \\
 &= \lim_{n \rightarrow \infty} \frac{\cos\left(\frac{b}{2n}\right) - \cos\left(b + \frac{b}{2n}\right)}{2 \sin\left(\frac{b}{2n}\right)} \cdot \frac{b}{n} \\
 &= \lim_{n \rightarrow \infty} \left(\cos\left(\frac{b}{2n}\right) - \left(\cos b \cos\left(\frac{b}{2n}\right) - \sin b \sin\left(\frac{b}{2n}\right) \right) \right) \times \frac{b}{2n} \times \frac{1}{\sin\left(\frac{b}{2n}\right)} \\
 &= \lim_{n \rightarrow \infty} \left(\cos\left(\frac{b}{2n}\right) - \cos b \cos\left(\frac{b}{2n}\right) + \sin b \sin\left(\frac{b}{2n}\right) \right) \times \lim_{n \rightarrow \infty} \frac{b}{2n} \times \frac{1}{\sin\left(\frac{b}{2n}\right)} \\
 &= (1 - \cos b + 0) \times 1 \quad \text{since } \lim_{n \rightarrow \infty} \frac{\theta}{\sin \theta} = 1, \text{ and } \lim_{n \rightarrow \infty} \cos \frac{b}{2n} = 1, \text{ and} \\
 &\quad \lim_{n \rightarrow \infty} \sin \frac{b}{2n} = 0 \quad \text{since } \lim_{n \rightarrow \infty} \frac{b}{2n} = 0.
 \end{aligned}$$

$= 1 - \cos b$
as required.

3 marks	Obtaining line marked (*) AND taking each of the two limits correctly to obtain the correct expression
2 marks	Obtaining line marked (*) AND taking one of the limits correctly
1 mark	Obtaining line marked (*)